Home Guardian

A self-learning Wi-Fi system that protects elders and empty homes. No camera. Nothing to wear.

Shravya Shetty | M.Tech CSE project
One-page summary | Nov 2026 (design stage)

What it is

Small, cheap **ESP32 Wi-Fi boards** (about Rs. 400-800 each) are placed around the home. When a person moves, the Wi-Fi signal between the boards changes a little. An **AI model** reads these tiny changes (called **CSI**: the fine details of a Wi-Fi signal) and understands what is happening: walking, sitting, lying, a fall, or a stranger walking in.
It **learns each home's own pattern** and keeps getting better every night.

Why it is useful

- About **684,000 people die from falls** every year; over 80% of them in low- and middle-income countries like India (WHO).
- Many elders live alone. After a fall they may lie on the floor for hours.
- Cameras feel like spying, and wearables get forgotten or are not charged. This needs **neither**.
- The **same boards also guard the empty house** against theft, so families get two products for the price of one.

What it does: two modes, one system

GUARDIAN MODE (family at home)
- Detects a **fall**, a **very long stillness** (e.g. 40 min in the bathroom) and an **unusual routine** (e.g. no movement by 9 am).
- Speaker asks **"Are you OK?"**. If there is no reply, it calls or sends an SMS to the family.

THEFT MODE (house empty or at night)
- Detects **any person moving** and tells **which floor and room**.
- Ignores pets, fans and curtains. Knows **family from stranger** (phones + walking style). If it's a stranger: alarm + phone alert.

Auto switch: when every family phone leaves the home Wi-Fi it switches to Theft mode; when a known phone comes back it switches to Guardian mode. Night mode can use both.

How it works

1. Sense Boards record the Wi-Fi signal 100 times a second → **2. Clean** Remove noise and sudden spikes → **3. AI brain** Turns each 2-sec clip into a "movement fingerprint" → **4. Three answers** What is happening? Is it normal for this home? Where and who? → **5. Double check** 2 boards must agree, then the voice check → **6. Alert + live map** Phone alert; live house map with moving dots

Self-learning: every night the model re-trains on the day's data. Each reply ("I'm OK", "only the cat") becomes a lesson, and old examples are replayed so it does not forget. Starts from a public ESP32 dataset (UIT-ESP32) + our own recorded data. Everything runs at home: **no cloud, no camera**.

How it is better than published work

	WiFall 2017	DeFall 2022	XFall 2024	SiFall 2022	AR-Alarm 2017	UIT-ESP32 2026	Home Guardian (ours)
Detects falls	✓	✓	✓	✓	–	✓	✓
Detects intruders / theft	–	–	–	–	✓	–	✓
Keeps learning the home's routine	–	–	–	✓*	–	–	✓
Knows family vs stranger	–	–	–	–	–	–	✓
"Are you OK?" voice check	–	–	–	–	–	–	✓
Cheap ESP32 boards	–	–	–	–	–	✓	✓
Which floor and room (2+ floors)	–	–	–	–	–	room	✓
Reported result	94% precision, 13% false alarms	>95% caught, <1.5% false	96.8% accuracy	94.1% caught, 1 false / 3 days	intruders only; see paper	>99% precision	Target: >=95%, <1 false / week

* SiFall self-learns for falls only. – = not reported in that paper. In our search of 11 papers, **no single system combines all seven rows**.

6 features that make it stand out

1. **Two modes, one system**: fall alert + theft alarm on the same boards.
2. **Learns the home's routine** by itself, every night, without labels.
3. **Family vs stranger** using phones + walking style: no false theft alarm when a son comes home early.
4. **"Are you OK?" voice check**: fewer false alarms, and every answer teaches the model.
5. **Cheap multi-floor setup** that tells floor and room.
6. **Live dashboard**: house map, current mode, confidence meter and alerts on a phone.

Cost and honest goal

Prototype for a 2-floor home: about Rs. 3,000-6,000
- 6 ESP32 boards (1 sender + 2 receivers per floor)
- A laptop as the hub now; a Raspberry Pi later
- Small speaker + mic (or an old phone)

Goal: catch at least 95% of falls and intruders with less than 1 false alarm a week after learning. No system is 100%; the voice check covers the rest.
Tests: acted falls on a mattress, 1-3 people, 2 floors, 2-4 weeks of live use, compared with WiFall-style and SiFall-style methods.

Build plan: 1. Live dashboard on simulated data → 2. Record Wi-Fi data with 3 ESP32 boards in one room → 3. Train the first model (falls + empty/motion) → 4. Add Theft mode, 2 floors and night learning → 5. Full tests and write the paper.

References: WHO, "Falls" fact sheet. | Wang, Wu, Ni, "WiFall", IEEE TMC 16(2), 2017. | Hu et al., "DeFall", IEEE IoT-J 9, 2022. | Chi et al., "XFall", IEEE JSAC 42(9), 2024. | Ji, Xie, Li, "SiFall", ACM SenSys 2022. | Li et al., "AR-Alarm", ICOST 2017, LNCS 10461. | Do et al., "Human Fall Detection and Indoor Localization Using WiFi Sensing" (UIT-ESP32), LNNS 1581, 2026. | Yang et al., "AutoFi", IEEE IoT-J 10(8), 2023. Full list with links: literature/notes/wifi-fall-detection.md

Paper figures are the authors' own reports. Home Guardian figures are targets, not results yet. Uniqueness is based on our search so far.